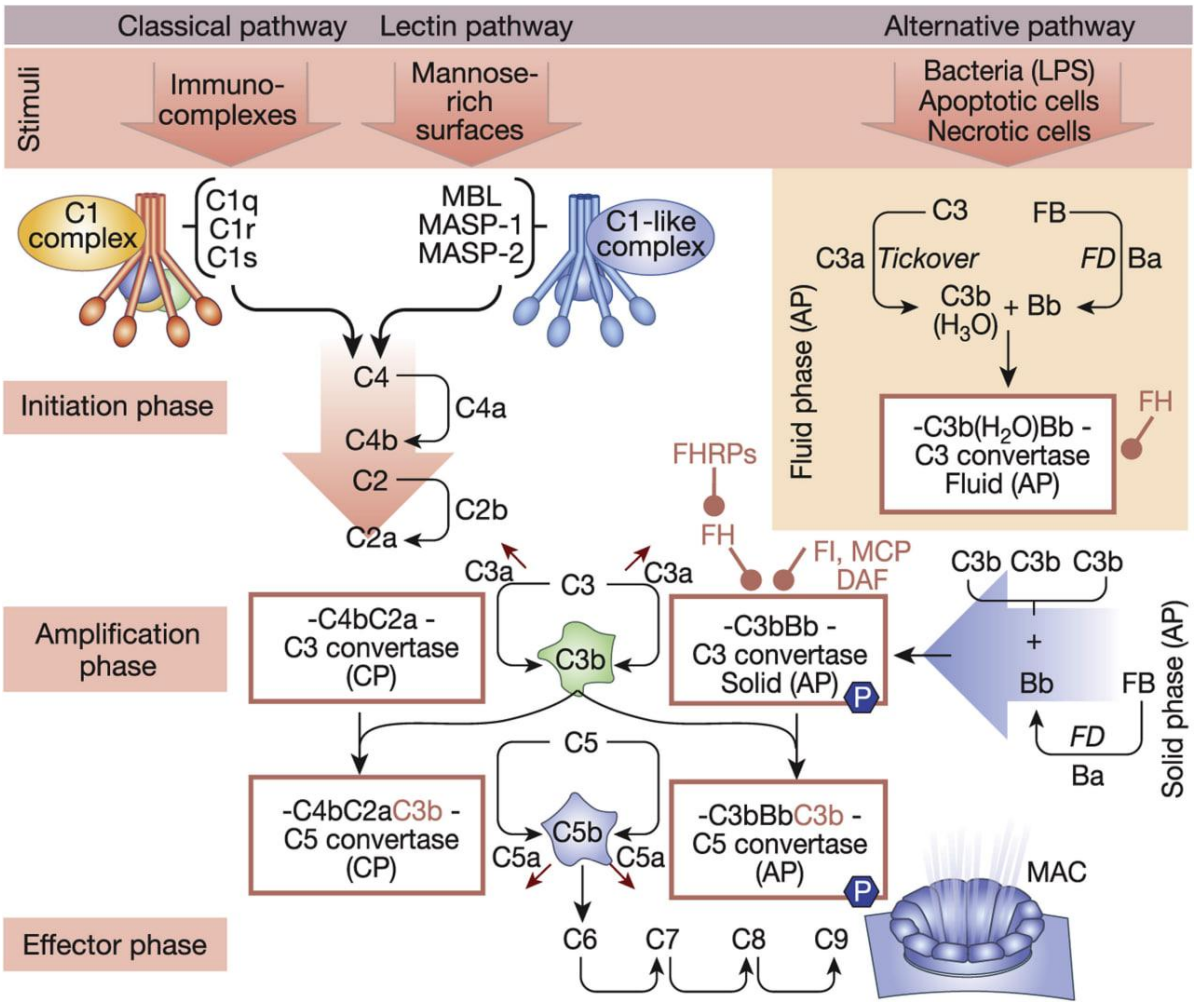


The figure uses **C2a = large fragment, C2b = small fragment**, so the convertase is written **C4bC2a**.

A. CLASSICAL PATHWAY (left column)

Stimuli: Immune complexes (antibody, IgG or IgM, bound to antigen on a microbe).

Initiation phase

1. **C1 complex** (orange oval): made of **C1q** + **C1r** + **C1s**.
    - **C1q** is the recognition part (six globular heads, like a bunch of tulips). Its heads bind the **Fc region** of the antibody.
    - **C1r and C1s** are serine proteases.
2. **Binding of C1q** changes the shape of the complex, so **C1r auto-activates** and then **activates C1s**.
3. **C1s cleaves C4:**
    - **C4a**: small fragment, released (arrow going right in the figure).

- **C4b**: large fragment, binds the microbial surface through its exposed thioester bond.

4. **C1s cleaves C2**, which has bound to C4b:
    - **C2b**: small fragment, released.
    - **C2a**: large fragment, stays on C4b.
5. **C4b + C2a = C4bC2a**, the **C3 convertase of the classical pathway** (red box labelled "CP").

B. LECTIN PATHWAY (middle column)

Stimuli: Mannose-rich surfaces: repeating mannose (also fucose, GlcNAc) on bacteria, yeasts and some viruses.

Initiation phase

1. **MBL** (mannose-binding lectin) binds the sugar arrays. It is associated with **MASP-1 and MASP-2** (MBL-associated serine proteases). Together they form the **"C1-like complex"** (blue), named so because MBL looks like C1q and the MASPs work like C1r/C1s.
2. Binding activates **MASP-1**, which activates **MASP-2**.
3. **MASP-2 cleaves C4** → C4a (released) + C4b (binds the surface).
4. **MASP-2 cleaves C2** → C2b (released) + C2a (stays).
5. Result: the **same C4bC2a C3 convertase** as the classical pathway.

Why the figure draws one shared stream: from C4 onward, classical and lectin pathways are identical. The only differences are the trigger and the initiating enzymes (C1r/C1s vs MASP-1/MASP-2).

C. ALTERNATIVE PATHWAY (right column)

Stimuli: Bacteria (LPS), apoptotic cells, necrotic cells. These are surfaces that lack the full set of complement regulators, so C3b is allowed to stay and amplify. (This pathway needs no antibody and no lectin.)

Fluid phase (top orange box): getting started in plasma

1. **Tick-over:** native **C3** slowly and spontaneously hydrolyses its thioester bond by reacting with water, giving **C3b(H_2O)** (really C3(H_2O), which acts like C3b). **C3a** is released.
2. **Factor B (FB)** binds C3b(H_2O).

3. **Factor D (FD)** cleaves factor B → **Ba** (small, released) + **Bb** (large, stays; the protease).
4. Result: **C3b(H_2O)Bb**, the **fluid-phase C3 convertase of the AP**.
5. **Factor H (FH)** (red dot = inhibition) limits this convertase in healthy plasma.
6. This convertase cleaves some C3, making the first real **C3b**, which can land on a surface.

Solid phase (blue arrow, right): on the microbial surface

1. Many **C3b** molecules deposit on the surface (the "C3b C3b C3b" at the top of the blue arrow).
2. Each C3b binds **factor B**, and **factor D** cleaves it into Ba + Bb.
3. Result: **C3bBb**, the **solid-phase (surface) C3 convertase of the AP** (red box labelled "Solid (AP)").
4. **Properdin (P**, the blue hexagon) **stabilises** C3bBb, extending its short half-life.

D. AMPLIFICATION PHASE (middle, where all three meet)

1. Both C3 convertases (**C4bC2a** from CP/LP, and **C3bBb** from AP) cleave **C3** into:
    - **C3a** (red arrows, released): anaphylatoxin, inflammation.
    - **C3b** (green shape in the centre): the hub of the figure.
2. **C3b has three jobs:**
    - **Opsonisation**: coats the microbe for phagocytes.
    - **Amplification loop**: new C3b binds factor B and makes more C3bBb (the arrow from C3b back to the AP convertase). This is why the AP produces most of the C3b in practice, even when the CP or LP started the response.
    - **Forms the C5 convertase** (next section).
3. **Regulators (red dots, inhibition):**
    - **FH** (factor H) and **FHRPs** (factor H-related proteins), **FI** (factor I), **MCP** (membrane cofactor protein), **DAF** (decay-accelerating factor).
    - These inactivate C3b and dissociate convertases on host cells, which is why amplification is restricted to microbes. (FHRPs are mainly thought to compete with factor H and can enhance complement activation, though your textbook's wording may differ.)

E. EFFECTOR PHASE (bottom)

1. **A further C3b joins the C3 convertase** (the red "C3b" in the labels), forming the **C5 convertase**:
    - **CP/LP: C4bC2aC3b**
    - **AP: C3bBbC3b**
2. The C5 convertase cleaves **C5**:
    - **C5a**: small, released. A strong anaphylatoxin and neutrophil chemoattractant (red arrows).
    - **C5b**: large, stays; starts the membrane attack complex.
3. **C5b** sequentially binds **C6, C7, C8, then many C9**.
4. This builds the **MAC (membrane attack complex, C5b-9)**, the pore (drawn at bottom right) that causes osmotic lysis of the microbe.

F. Comparison table from the figure

	Classical	***Lectin***	***Alternative***
Stimulus	Immune complexes	Mannose-rich surfaces	LPS, apoptotic/necrotic cells
Initiator	C1q (+ C1r, C1s)	MBL (+ MASP-1, MASP-2)	Tick-over $C3b(H_2O)$ + FB + FD
Needs antibody	Yes	No	No
C3 convertase	C4bC2a	C4bC2a	C3bBb (stabilised by properdin)
C5 convertase	C4bC2aC3b	C4bC2aC3b	C3bBbC3b
Small fragments released	C4a, C2b	C4a, C2b	Ba
Ends in	C5 → MAC	C5 → MAC	C5 → MAC

G. Fragments that stay vs fragments released

- **Stay and do the work:** C4b, C2a, C3b, Bb, C5b
- **Released:** C4a, C2b, Ba, C3a, C5a (the "a" fragments of C3, C4 and C5 signal inflammation)

One-sentence summary: three different triggers build one of two C3 convertases (C4bC2a or C3bBb), which make C3b; C3b amplifies through the AP loop and builds the C5 convertase, which cleaves C5 so that C5b assembles the MAC while C3a and C5a drive inflammation.

Biological functions

Function	*Mediators*	*Mechanism*
Opsonisation	C3b, iC3b (inactivated C3b), C4b	Bind **complement receptor 1 (CR1, CD35)** and **CR3 (CD11b/CD18)** on phagocytes, promoting phagocytosis
Inflammation	**C3a, C4a, C5a** (**anaphylatoxins**)	Bind C3aR and C5aR on mast cells and basophils → degranulation (histamine), vasodilation, increased vascular permeability; C5a also attracts and activates neutrophils
Lysis	MAC (C5b-9)	Pore formation in the target membrane
Immune complex clearance	C3b, CR1 on red cells	Red cells carry complexes to the spleen and liver for removal
Apoptotic cell clearance	C1q, MBL, C3b	Tag dying cells for phagocytosis
B cell activation	C3d binding **complement receptor 2 (CR2, CD21)**	Lowers the B cell activation threshold, boosting antibody responses

C5a is the most potent anaphylatoxin and the main chemoattractant. Anaphylatoxins are quickly inactivated by **carboxypeptidase N**.

Regulation of complement

Needed to prevent self-damage.

Regulator	*Full name*	*Action*
C1-INH	C1 inhibitor	Inhibits C1r, C1s, **MASP-1 and MASP-2**; also inhibits kallikrein and factor XIIa
C4BP	C4-binding protein	Accelerates decay of C4b2a; cofactor for factor I

Factor H		Competes with factor B for C3b; accelerates decay of C3bBb; cofactor for factor I
Factor I		Cleaves C3b and C4b (needs cofactors) into inactive forms
DAF (CD55)	Decay-accelerating factor	Dissociates C3 convertases on host cells
MCP (CD46)	Membrane cofactor protein	Cofactor for factor I
CR1 (CD35)	Complement receptor 1	Decay accelerator and cofactor
CD59	Protectin	Blocks C9 binding, preventing MAC assembly
S protein (vitronectin), clusterin		Prevent MAC insertion in plasm

Complement deficiencies

Deficiency	***Consequence***
C1q, C1r/s, C2, C4	Immune complex disease, **SLE**-like illness (failure to clear complexes and apoptotic cells)
C3	Severe recurrent pyogenic (pus-forming) infections; immune complex glomerulonephritis
Factor H, factor I	Uncontrolled C3 consumption → low C3, recurrent infections; factor H deficiency also causes **atypical haemolytic uraemic syndrome (aHUS)**
Properdin *(X-linked)*	Recurrent *Neisseria* infections
C5, C6, C7, C8, C9 *(terminal)*	Recurrent ***Neisseria*** infections (meningococcal meningitis, gonorrhoea)
C1-INH	**Hereditary angioedema (HAE)**: episodes of swelling from excess **bradykinin**; low C4 even between attacks
DAF / CD59 *(acquired, PIGA mutation)*	**Paroxysmal nocturnal haemoglobinuria (PNH)**: complement-mediated haemolysis of red cells